SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT TOOL 2 – Pseudocode Writing Task (Algorithm Design)

Course Code:	DSA0307	**Course Title:**	Natural Language Processing
CO Assessed:	CO5 – Articulation	**Assessment:**	Assessment Tool 2 – Pseudocode Writing Task (Algorithm Design)
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	AI & ML / 2024	**Date:**	18/08/2026

Code of Conduct

I _D. Sri Anjaneya / 192425219_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __D. Sri Anjaneya__

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Reference Resolution, Discourse	Entity Linking Algorithm	Q1
Text Coherence, Discourse Structure	Coherence Analysis Algorithm	Q2
Dialog Acts, Conversational Agents	Intent Recognition Algorithm	Q3
Language Generation, Surface Realization	NLG Pipeline Design	Q4
Machine Translation, Interlingua, Statistical Approaches	Translation Workflow Algorithm	Q5

Annexure A – Pseudocode Writing Task (Algorithm Design)

1. Design an algorithm to perform **Reference Resolution** by identifying pronouns and linking them to the correct entities in a discourse. Apply your algorithm to the text:

"Ravi met Arun at the library. He borrowed a book and later returned it."

Generate the resolved discourse by replacing pronouns with their corresponding entities. Further, validate the correctness of the resolution process by explaining how discourse context is used to identify the antecedents.

2. Develop an algorithm to analyze **Text Coherence** and **Discourse Structure** by identifying logical relationships between sentences such as cause-effect, sequence, and elaboration. Apply your algorithm to the following discourse:

"The roads were flooded after heavy rainfall. Therefore, schools were closed for the day. Students attended classes online."

Identify the discourse relations among the sentences and construct a discourse structure representation. Further, justify how these relations contribute to maintaining coherence in the text.

3. Design an algorithm for a **Conversational Agent** that identifies **Dialogue Acts** from user interactions. Apply your algorithm to the conversation:

User: "Can you book a train ticket for me?"
Agent: "Sure, where would you like to travel?"
User: "I want to go to Chennai."
Agent: "Your ticket has been booked."

Classify each utterance as Request, Question, Inform, Confirmation, or Action. Generate the dialogue-act sequence and evaluate how the identified acts help the conversational agent interpret user intentions.

4. Design an algorithm for **Language Generation** that converts a semantic representation into a grammatically correct sentence through **Surface Realization**. Consider the semantic input:

```
Action: Buy
Agent: Student
Object: Book
Tense: Past
```

Generate the final natural language sentence and explain how the algorithm performs lexical selection, sentence structuring, and surface realization. Further, validate the grammatical correctness of the generated output.

5. Design an algorithm that combines **Interlingua-based Machine Translation** and **Statistical Translation Approaches**. Apply your algorithm to translate the sentence:

"The boy is playing football."

Perform the following steps:

1. Analyze the source sentence.
2. Create an Interlingua representation.

3. Generate candidate target-language translations.
4. Apply statistical scoring to select the most appropriate translation.
5. Produce the final translated sentence.

Finally, evaluate how the Interlingua representation and statistical methods improve translation quality and reduce ambiguity.

Q1. Reference Resolution – Entity Linking Algorithm

Problem Understanding

Input: A discourse containing entities and pronouns.

Output: Pronouns linked to their correct antecedents and a resolved discourse.

Constraints: Gender, number, recency, grammatical compatibility, and discourse context.

Pseudocode

ALGORITHM Reference_Resolution(Text)

INPUT: Text containing entities and pronouns

OUTPUT: Resolved text with pronouns replaced by antecedents

1. Split Text into sentences.
2. Identify all named entities and noun phrases.
3. Identify all pronouns in each sentence.
4. FOR each pronoun P:

    Create a candidate list of possible antecedents.

    FOR each candidate C:

      IF gender(P) matches gender(C)

        AND number(P) matches number(C)

        AND C occurs before P

      THEN

        Calculate Score(C):

          +3 for gender agreement

          +3 for number agreement

          +2 for grammatical compatibility

          +2 for recent occurrence

          +2 for discourse coherence

      ELSE

        Reject C.

    Select candidate with highest Score.

    Link P to selected antecedent.
5. Replace each pronoun with its selected antecedent.
6. Return resolved text.

END ALGORITHM

Application

Given:

“Ravi met Arun at the library. He borrowed a book and later returned it.”

Entities:

- Ravi
- Arun
- book

Pronouns:

- **He**
- **it**

For **He**, possible antecedents are Ravi and Arun. Both are masculine, but **Arun is more recent**, so the algorithm selects Arun.

For **it**, the most semantically compatible antecedent is **book**, because a person does not normally get “returned” in this context, while a book can be returned.

Coreference Chain

Ravi ——————————————→ Ravi

Arun ——→ He ——————————→ Arun

Book ——→ it ——————————→ Book

Resolved Discourse

Ravi met Arun at the library. Arun borrowed a book and later returned the book.

Validation

The resolution is correct because **He** agrees in gender and number with both Ravi and Arun, but **Arun is more recent**, making it the preferred antecedent. The pronoun **it** is linked to **book** because of semantic compatibility with the verb **returned**. Thus, grammatical agreement, recency, semantics, and discourse coherence collectively support the final resolution.

Q2. Text Coherence and Discourse Structure Algorithm

Problem Understanding

Input: Multiple related sentences.

Output: Discourse relations and a coherent discourse structure.

Relations: Cause-Effect, Sequence, Elaboration.

Pseudocode

ALGORITHM Coherence_Analysis(Text)

INPUT: Set of sentences S1, S2, ..., Sn

OUTPUT: Discourse relations and discourse structure

1. Split Text into sentences.
2. FOR each pair of consecutive sentences (Si, Si+1):
    Identify keywords such as:
        because, therefore, after, then, additionally, etc.

    Identify semantic relationships between sentences.

    IF Si causes Si+1
        Relation = CAUSE-EFFECT
    ELSE IF Si occurs before Si+1
        Relation = SEQUENCE
    ELSE IF Si provides more information about Si-1
        Relation = ELABORATION
    ELSE
        Relation = OTHER
3. Construct a discourse tree using identified relations.
4. Check whether each sentence is logically connected.
5. Return relations and discourse tree.

END ALGORITHM

Application

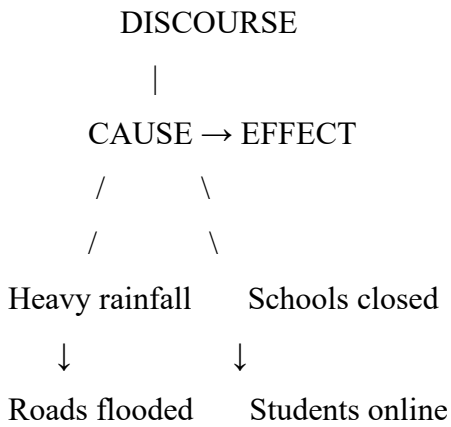
Given:

“The roads were flooded after heavy rainfall. Therefore, schools were closed for the day. Students attended classes online.”

Discourse Relations

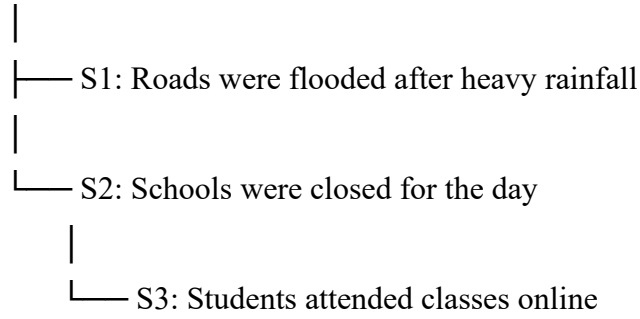
- **Sentence 1 → Sentence 2:** Cause-Effect
    - Flooded roads → schools closed.
- **Sentence 2 → Sentence 3:** Cause-Effect / Result
    - Schools closed → students attended classes online.

Discourse Structure



More formally:

ROOT: CAUSE-EFFECT



Validation

The text is coherent because each sentence logically follows from the previous event:

Heavy rainfall

↓

Roads flooded

↓

Schools closed

↓

Students attended online

The word **"Therefore"** explicitly signals the cause-effect relation between S1 and S2. S3 provides the logical consequence of school closure. Hence, the discourse forms a continuous and meaningful event chain.

Q3. Dialogue Act / Intent Recognition Algorithm

Problem Understanding

Input: User and agent utterances.

Output: Dialogue-act label for every utterance and complete dialogue sequence.

Pseudocode

ALGORITHM Intent_Recognition(Utterance)

INPUT: Conversation utterance

OUTPUT: Dialogue Act

1. Read the utterance.
2. IF utterance asks the system to perform an action
   RETURN REQUEST
3. ELSE IF utterance asks for information
   RETURN QUESTION
4. ELSE IF utterance provides information
   RETURN INFORM
5. ELSE IF utterance confirms a previous action
   RETURN CONFIRMATION
6. ELSE IF utterance performs or reports an action
   RETURN ACTION
7. ELSE
   RETURN UNKNOWN

END ALGORITHM

Application

Conversation:

Speaker	**Utterance**	**Dialogue Act**	**Reason**
User	"Can you book a train ticket for me?"	**Request**	User requests an action
Agent	"Sure, where would you like to travel?"	**Question**	Requests missing information
User	"I want to go to Chennai."	**Inform**	Provides destination information
Agent	"Your ticket has been booked."	**Action**	Reports completion of booking

Dialogue-Act Sequence

Request → Question → Inform → Action

Intent Flow

User wants ticket

↓

REQUEST

↓

Agent needs destination

↓

QUESTION

↓

User provides Chennai

↓

INFORM

↓

Agent completes booking

↓

ACTION

Evaluation

The dialogue acts allow the conversational agent to understand the user's intention step by step. The **Request** identifies the main goal, the **Question** obtains missing information, the **Inform** supplies the destination, and the **Action** completes the requested task. Therefore, dialogue-act recognition maintains conversational coherence and enables correct task execution.

Q4. Natural Language Generation / Surface Realization Algorithm

Semantic Input

Action = Buy

Agent = Student

Object = Book

Tense = Past

Problem Understanding

Input: Semantic representation.

Output: Grammatically correct natural-language sentence.

Main operations: Lexical selection, sentence structuring, surface realization.

Pseudocode

ALGORITHM NLG_Surface_Realization(Semantic_Input)

INPUT: Action, Agent, Object, Tense

OUTPUT: Natural language sentence

1. Read semantic representation.
2. Select lexical items:
   - Agent → "student"
   - Action → "buy"
   - Object → "book"
3. Determine sentence structure:
   - Subject + Verb + Object
4. Check Tense.
5. IF Tense = Past
   - Change "buy" → "bought"
6. Add required determiners:
   - "student" → "The student"
   - "book" → "a book"
7. Arrange words:
   - The student + bought + a book.
8. Apply capitalization and punctuation.
9. Validate grammar.
10. Return final sentence.

END ALGORITHM

Surface Realization

Semantic Representation

↓

Lexical Selection

↓

Student → "the student"

Buy → "bought"

Book → "a book"

↓

Sentence Structure

↓

Subject + Verb + Object

↓

Surface Realization

↓

"The student bought a book."

Final Sentence

The student bought a book.

Validation

The sentence is grammatically correct because:

- **The student** = Subject
- **bought** = Past-tense verb
- **a book** = Object
- Word order = **Subject + Verb + Object**
- The sentence begins with a capital letter and ends with a full stop.

Thus, the generated sentence correctly preserves all semantic information.

Q5. Interlingua + Statistical Machine Translation Algorithm

Source Sentence

“The boy is playing football.”

Problem Understanding

Input: English source sentence.

Output: Appropriate target-language translation.

Methods: Interlingua representation + candidate generation + statistical scoring.

For demonstration, the target language is **Tamil**.

Pseudocode

ALGORITHM Hybrid_Translation(Source_Sentence)

INPUT: Source sentence

OUTPUT: Best target-language translation

1. Analyze Source_Sentence.
2. Perform:
    Tokenization
    Part-of-speech analysis
    Semantic analysis
3. Extract:
    Subject = boy
    Action = play
    Object = football
    Tense/Aspect = Present Continuous
4. Create Interlingua representation.
5. Generate multiple target-language candidates.
6. FOR each candidate:
    Calculate statistical score using:
        Translation Model Score
        Language Model Score
        Semantic Consistency Score
7. Select candidate with highest total score.
8. Perform grammatical surface realization.
9. Return final translation.

END ALGORITHM

Step 1: Source Sentence Analysis

Word	Role
The boy	Subject
is playing	Present continuous action
football	Object

Semantic meaning:

Agent = Boy

Action = Play

Object = Football

Aspect = Present Continuous

Step 2: Interlingua Representation

The Interlingua is language-independent.

```
PLAY(
   AGENT = BOY,
   OBJECT = FOOTBALL,
   ASPECT = PRESENT_CONTINUOUS
)
```

Or:

```
[EVENT: PLAY]
   |
   ├── AGENT: BOY
   ├── OBJECT: FOOTBALL
   └── ASPECT: CONTINUOUS
```

The important meaning is represented without depending on English grammar.

Step 3: Generate Candidate Tamil Translations

Possible candidates:

C1 = சிறுவன் கால்பந்து விளையாடுகிறான்.

C2 = பையன் கால்பந்து விளையாடுகிறான்.

C3 = சிறுவன் கால்பந்து விளையாடினார்.

Their meanings are approximately:

- **C1:** The boy is playing football.
- **C2:** The boy is playing football.
- **C3:** The boy played football / is expressed in a past-oriented form.

Step 4: Statistical Scoring

The system evaluates each candidate using translation probability, language-model probability, and semantic consistency.

Example conceptual scoring:

Candidate	Translation Score	Language Score	Semantic Score	Total
C1	0.95	0.94	0.98	**0.96**
C2	0.92	0.93	0.97	**0.94**
C3	0.70	0.85	0.60	**0.72**

Therefore:

Highest Score → C1

Step 5: Final Translation

சிறுவன் கால்பந்து விளையாடுகிறான்.

Meaning:

The boy is playing football.

How Interlingua Improves Translation

Interlingua separates **meaning from language-specific surface form**.

English

"The boy is playing football."

↓

INTERLINGUA

↓

Agent = Boy

Action = Play

Object = Football

Aspect = Continuous

↓

Tamil

"சிறுவன் கால்பந்து விளையாடுகிறான்."

How Statistical Methods Improve Translation

Statistical scoring helps select the most probable and natural target sentence from multiple candidates.

It considers:

- Translation probability
- Target-language fluency
- Context
- Semantic consistency
- Grammatical likelihood

Rubrics for Calculation

Criteria	Weigh tage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understandi ng	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completene ss	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structu res	Pseudoc ode Structur e	Complete ness	Sub. Total	Total %
1							
2							
3							

4							
5							

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____